# Marçal Herraiz Bayó

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## Education

**MSc in Mathematics – PSL University** — Paris, France

*Rank: 5/116 (top 5%); GPA: 15.052/20; Minor in Physics* — Sept. 2025 – Present

- Coursework at Dauphine and École Normale Supérieure. Full FSMP scholarship.
- First-year thesis: "Market Making and Optimal Inventory Management" (18/20).

**BSc in Mathematics – University of Barcelona** — Barcelona, Spain

*Rank: 5/141 (top 4%); GPA: 9.02/10* — Sept. 2020 – June 2025

- Thesis: "The Riemann–Hilbert Correspondence for Flat Connections on Principal Bundles" (10/10).

**BSc in Physics – University of Barcelona** — Barcelona, Spain

*Rank: 16/196 (top 8%); GPA: 8.79/10* — Sept. 2020 – June 2025

- Thesis: "Improving Classical Shadows with Grouping Strategies" (9.7/10).

## Experience

**Massachusetts Institute of Technology (MIT)** — Cambridge, MA, USA

*Incoming Visiting Student – Particle Physics* — Apr. 2027 – Oct. 2027

- Will study Higgs boson decays into two muons using data from the CMS detector at CERN.
  *Supervisor: Prof. Dr. Christoph Paus.*

**Westinghouse Nuclear** — Madrid, Spain

*Simulation Modeling Engineer – Nuclear Plant Simulation* — July 2025 – Present

- Developing a compressible liquid–vapor–air model in Fortran with revised state variables and pressure solver.
- Secured leadership approval for a new multiphase-modeling project by presenting preliminary results.
- Built and validated a condenser heat-transfer model with under 2% error against plant data.
- Resolved major energy-conservation errors in open tanks by implementing a new overflow model.
- Delivered thermal-hydraulic models to production software for projects and plants under construction.

**Barcelona Supercomputing Center (BSC)** — Barcelona, Spain

*Visiting Student – Quantum Computing* — Nov. 2024 – June 2025

- Implemented a hybrid algorithm combining classical shadows with observable grouping using PennyLane.
- Improved energy-estimation accuracy up to 18-fold over standard classical shadows.
- Reduced measurement requirements by a factor of 100–1,000 compared with standard classical shadows.
  *Supervisors: Prof. Dr. Bruno Julià, Dr. Berta Casas, Dr. Sergi Masot.*

**Institute for Research in Biomedicine (IRB)** — Barcelona, Spain

*Research Intern – Computational Biophysics* — July 2024 – Feb. 2025

- Modeled DNA flexibility using coarse-grained computational methods in C.
- Extended the model to analyze how DNA damage affects flexibility.
- Developed Python algorithms to compute DNA geometric descriptors.
  *Supervisor: Prof. Dr. Modesto Orozco.*

**Center for Mathematical Research (CRM)** — Barcelona, Spain

*Research Intern – Complex Dynamics* — June – Aug. 2023

- Studied bifurcations in the complex exponential family and the non-structural stability of $\exp(z)$.
  *Supervisors: Prof. Dr. Núria Fagella, Prof. Dr. Xavier Jarque.*

## Honors and Awards

**Fondation Sciences Mathématiques de Paris**: full master's scholarship; 1,000+ applicants. 2025
**Excellent with Honors**: awarded in 16 subjects across both BSc programs at UB. 2021–2025
**Collaboration Scholarship**: research toward the Mathematics BSc thesis. 2024
**Maths4Life Award**: one of five IRB grants for research in mathematics and biomedicine. 2024
**Honors in Baccalaureate**: academic distinction covering first-year university tuition in full. 2020
**Mad for Science Programme**: selective Physics research programme for high-school students. 2019

## Skills

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|---|---|
| **Programming** | Proficient: Python (NumPy, SciPy, pandas, scikit-learn), Fortran, C; familiar: R, Bash. |
| **Numerics** | Monte Carlo & variance reduction, ODE/SDE solvers, finite differences, particle methods. |
| **Inference & ML** | Regression, regularization, Bayesian inference, PCA, SVMs, NNs, graphical models, time series. |
| **Development** | Debugging, model validation, legacy-code refactoring, code review, technical documentation. |
| **Software** | Git/GitHub, Unix/Linux, Jupyter, PennyLane, LaTeX, Mathematica, SageMath. |
| **Languages** | Spanish and Catalan (native) • English (C1) • French (A2). |